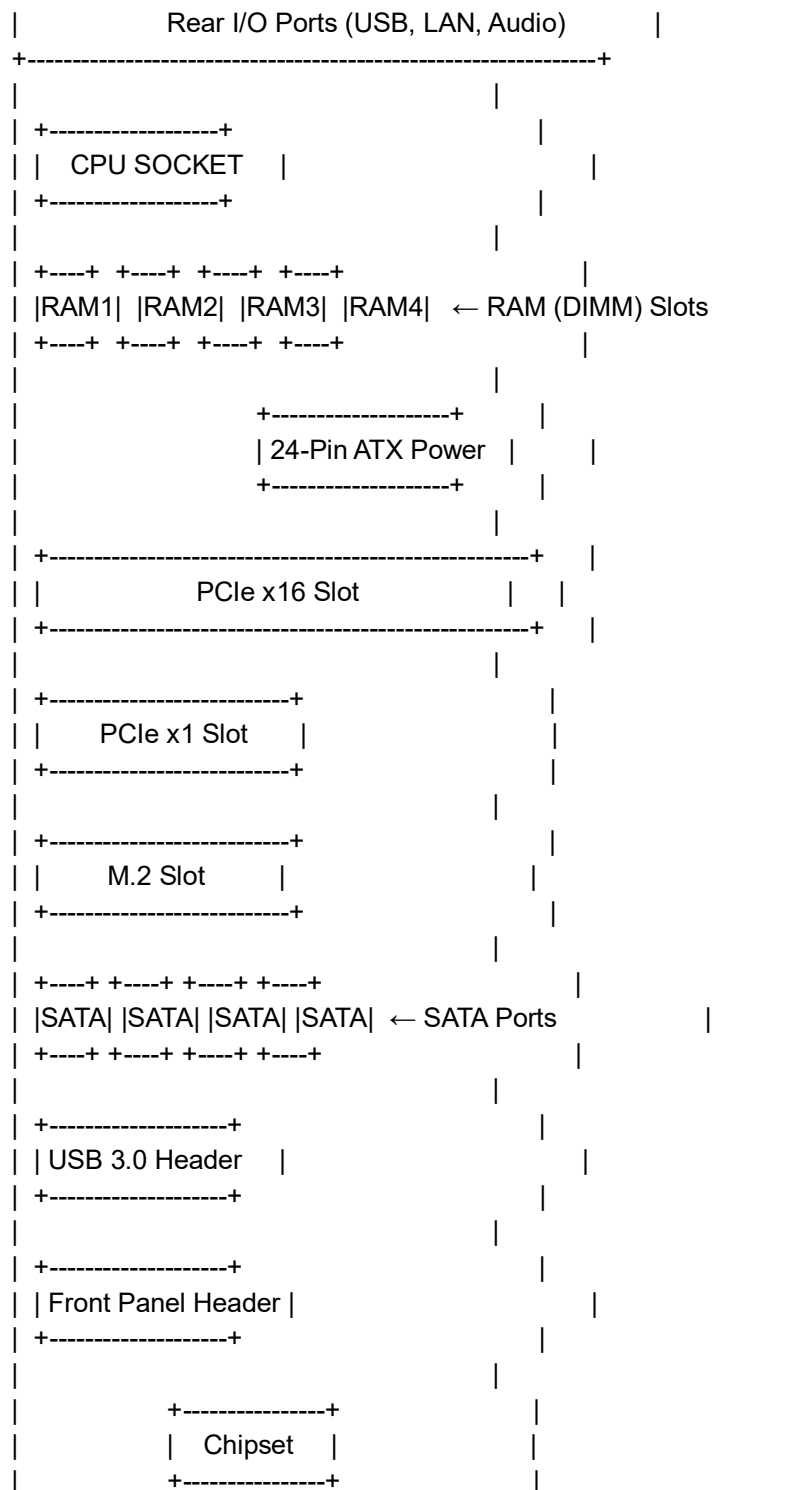


MOTHERBOARD AND CPU THEORY

- 1) Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.)

❖ Labeled Diagram of an ATX Motherboard



❖ **Marked Slots/Ports**

- CPU Socket
- RAM (DIMM) Slots
- PCIe x16 Slot
- PCIe x4 Slot
- M.2 Slot
- SATA Ports
- 24-Pin ATX Power Connector
- USB 3.0 Header
- Front Panel Header
- Rear I/O Ports

2) Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

❖ **POPULER SOCKET TYPES**

Socket Type	Manufacturer	Compatible CPU Example
LGA1151	Intel	Intel Core i7-9700K
AM4	AMD	AMD Ryzen 5 5600X
LGA1200	Intel	Intel Core i5-10400

3) Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

❖ **CPU (Central Processing Unit)**

- The CPU is the brain of the computer. It performs calculations, executes program instructions, and manages tasks requested by software.

❖ Chipset

- The chipset is a controller on the motherboard that manages communication between the CPU and other hardware devices such as USB ports, SATA drives, network adapters, and audio devices.

❖ Example: Opening wynk music

- You click the wynk music icon.
- The CPU processes the command and starts the application.
- The chipset helps transfer data from the SSD to RAM.
- The CPU loads wynk's program files into memory.
- The chipset manages communication with audio hardware and network devices.
- wynk connects to the internet and streams music.
- The CPU processes the application while the chipset coordinates data flow between components.

- 4) Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labeling each one

